

SYS // ASL-NEURAL v1.0

REAL-TIME SIGN LANGUAGE DETECTION

Real-Time American Sign Language Detection

MediaPipe Hand Localization × ResNet-50 Classification

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// STATUS: ONLINE MODEL: RESNET-50 BACKEND: FASTAPI ACCURACY: 99.80%

Problem Statement

// THE BARRIER

- 70M+ deaf/hard-of-hearing individuals worldwide rely on sign language
- No real-time automated translation tools exist for everyday communication
- Sensor-based approaches (data gloves, depth cameras) impose hardware constraints
- Existing CNN systems lack real-time, low-latency feedback loops

// OUR SOLUTION

- **Two-stage real-time pipeline**

MediaPipe hand localization + ResNet-50 classification

- Live webcam integration with skeleton overlay
- FastAPI inference server + browser frontend
- **99.80% test accuracy in only 3 training epochs**

Data & Preprocessing

87K

LABELED IMAGES

29

CLASSES (A-Z + 3)

200×200

PX INPUT SIZE

// DATASET SPLITS

TRAIN

70%

~60,900 samples

VAL

20%

~17,400 samples

TEST

10%

~8,700 samples

// AUGMENTATION & TRANSFORMS

Training

- Random horizontal flip + color jitter
- Resize → 224×224 (ImageNet standard)

Evaluation

- Deterministic resize + normalize only

System Pipeline



// HAND DETECTOR (MEDIAPIPE)

- VIDEO mode – Kalman filter tracking
- Detection confidence threshold: 0.65 (vs default 0.50)
- 35% bbox expansion + exponential smoothing ($\alpha=0.7$)
- Motion gate: $L2 > 8px \rightarrow$ skip classification

// CLASSIFIER HEAD (SLD MODEL)

- ResNet-50 backbone – all params frozen
- 2048-dim $\rightarrow$ LayerNorm $\rightarrow$ GELU $\rightarrow$ Dropout(0.3) $\rightarrow$ 512 $\rightarrow$ 29
- LayerNorm vs BatchNorm: batch-size invariant, stable with frozen backbone
- Confidence threshold: 0.50 before acceptance

Training Configuration

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// HYPERPARAMETERS
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Optimizer	SGD (Nesterov)
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Learning Rate	1×10^{-2}
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Momentum	0.9
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Weight Decay	5×10^{-4}
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Scheduler	Cosine + Warm-Up
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Warm-Up Epochs	10
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Batch Size	128
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Grad Clip Norm	1.0
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Early Stopping	Patience = 20
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// LR SCHEDULE
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Warm-up phase (epochs $0 \rightarrow T_w$):

$$\eta_t = \eta_{\text{base}} \cdot (t + 1) / T_w$$

Cosine decay (post warm-up):

$$\eta_t = \eta_{\text{min}} + \frac{1}{2}(\eta_{\text{base}} - \eta_{\text{min}})(1 + \cos(\pi t))$$

```
// GRADIENT CLIPPING  $\tau = 1.0$ 
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$$\mathbf{g} \leftarrow \mathbf{g} \cdot \min(1, \tau / \|\mathbf{g}\|_2)$$

Prevents exploding gradients during early steep optimization phase

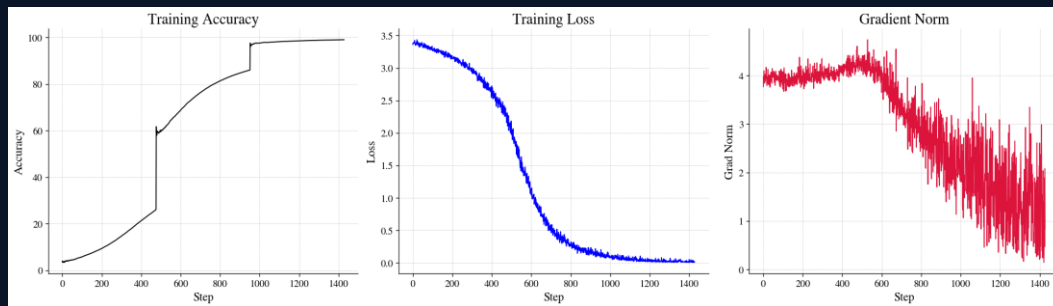
⚠ Trained for 3 epochs (Colab T4 session limit). Configured for 200 max epochs.

Training Results

// EPOCH METRICS

EPOCH	TRAIN ACC	VAL ACC
1 / 3	26.09%	64.41%
2 / 3	86.05%	98.29%
3 / 3	99.03%	99.86%

// W&B TRAINING CURVES



99.80%
TEST ACCURACY

Mean grad norm: 1.64 · Clip ratio: 0.77 (epoch 3)
Optimization still steep at termination – full run would yield marginal further gains

Real-Time Inference Pipeline

// INFERENCE STEPS

- | | | |
|---|-----------------------|--------------------------------|
| ① | Hand Detection | MediaPipe processes each frame |
| ② | Stability Gate | L2 motion > 8px → skip |
| ③ | Crop & Pad | Square ROI, edge replication |
| ④ | Classify | SLD model, conf ≥ 0.50 |
| ⑤ | Voting | 70% of 10-frame buffer |
| ⑥ | Cooldown | 1s per letter emission |

// BROWSER FRONTEND

- Client-side MediaPipe @ 60fps for zero-latency skeleton overlay
- Frames sent to backend @ ~8fps for classification
- Dwell gate: 1.8s hold to emit, 3.6s for repeats

System Demo – Live Feed

// ASL-NEURAL v1.0 – SIGN DETECT – MODEL: RESNET-50 BACKEND: ONLINE FPS: 7

SYS // ASL-NEURAL v1.0

SIGN DETECT

MODEL RESNET-50
BACKEND ONLINE
FPS 7

// SIGNAL STATUS

- HAND DETECTED
- STABLE
- CLASSIFYING...

// CONFIDENCE

CLF SCORE 0.30
VOTE BUFFER 0/10

// DWELL TIMER

0%
HOLD STILL
TO REGISTER
LETTER

// OUTPUT

HELLO

SPACE DEL CLR



// ALPHABET MATRIX

A	B	C	D	E	F	G
H	I	J	K	L	M	N
O	P	Q	R	S	T	U
V	W	X	Y	Z	del	spc
-						

// EVENT STREAM

```
> EMIT: O conf=99%
> EMIT: L conf=69%
> EMIT: L conf=73%
> EMIT: E conf=95%
> EMIT: H conf=100%
> EMIT: A conf=89%
> EMIT: E conf=91%
> EMIT: L conf=88%
> EMIT: O (*2) conf=99%
> EMIT: O conf=99%
> EMIT: B conf=95%
> EMIT: C conf=73%
> Camera online. System aimed.
> Repeat threshold: 500ms
> Dwell threshold: 1000ms
> Backend: http://localhost:8000
> System initialized.
```

HAND DETECTED

STABLE

CLASSIFYING

OUTPUT: HELLO

CONF: 99%

Conclusion & Future Directions

// ACHIEVEMENTS

- **99.80% test accuracy in just 3 training epochs**
- Real-time inference via FastAPI + browser frontend
- Dwell-based temporal voting for stable letter emission
- Full modular pipeline: detect → crop → classify → vote

// FUTURE DIRECTIONS

- Fine-tune MediaPipe for dedicated hand localization
- Upgrade to ResNet-152 or EfficientNet backbone
- Extend to dynamic gestures – LSTM / Transformer models
- **Mobile deployment for broader accessibility**

99.80%
TEST ACCURACY

3
EPOCHS ONLY

~8 FPS
REAL-TIME

Meaningful accessibility
technology – baseline for
full ASL word recognition